How the Digestive System Works

08/06/2026

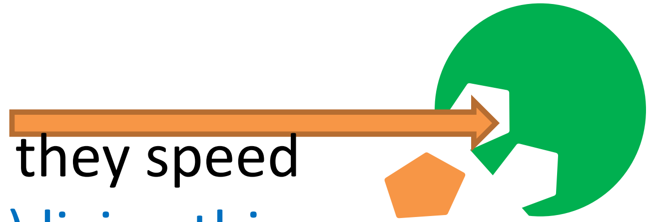
Keywords: Carbohydrase, amylase, protease, amino acid, lipase, glycerol, fatty acid, soluble, absorb

Starter: Complete these quiz questions about last lesson

Quiz 5: Factor Affecting Enzymes

1. Which enzyme is produced in the stomach a) Pepsin b) Amylase

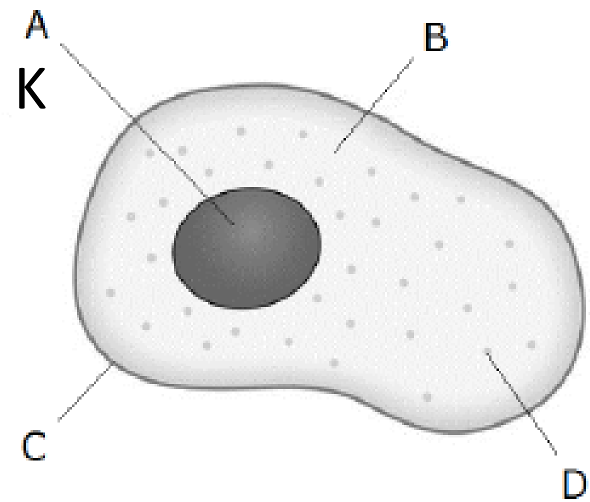
2. Enzymes are made of what?



3. Enzymes are biological catalysts which means they speed up reactions in a) dead things b) natural things c) living things

4. What is the arrow pointing to A________ s_____

5. What is the orange substance that joins with the enzyme for a reaction a) Substance b) Subject c) Sublet d) Substrate



6. What is the theory for how enzymes work L and K theory

7. What is part C

8. Name two things found in plant cells not found in animal cells V_______ and Chloro_______

9. Diffusion is the movement of molecules from a _______ concentration to a _______ concentration

10. Active transport and osmosis happens across a semi ______ membrane

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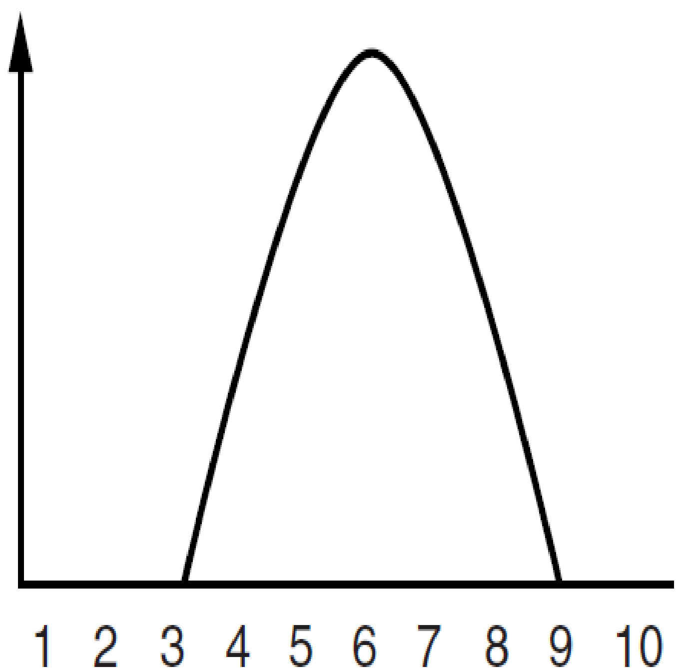
Keywords: Carbohydrase, amylase, protease, amino acid, lipase, glycerol, fatty acid, soluble, absorb

Starter: Answer these three review questions below.

EXT: What factors would affect how well the enzyme will work?

1. What are the missing graph labels? [2 marks]

2. Explain denaturing of enzymes and how it causes the rate of reaction to decrease. [3]



3. Explain why we get a temperature when we have an infection and why we try to decrease it. [4]

Review

1. What are the missing graph labels? [2 marks]

x axis = pH [1]
y axis = rate of reaction [1]

2. Explain denaturing of enzymes and how it causes the rate of reaction to decrease. [3]

*A process where **proteins lose their shape** due to pH or temperature limiting. It stops an enzyme being an effective catalyst because the **active site changes** shape and substrates can't fit.*

3. Explain why we get a temperature when we have an infection and why we try to decrease it. [4]

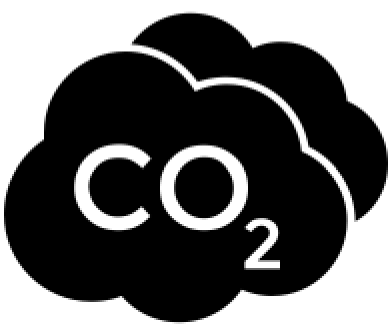
Some microorganisms can't reproduce at a higher temperature [1] Stops the infection spreading [1] High temperature can denature our enzymes [1] and we can die [1] so we try to reduce it.

LOb: Understand how the digestive system uses specific enzymes to absorb nutrients.

Keywords: Carbohydrase, amylase, protease, amino acid, lipase, glycerol, fatty acid, soluble, absorb

Learning Outcomes:

- Recall the digestive enzymes, where they are produced and what their substrate and product is
- Investigate the effect of pH on amylase enzyme



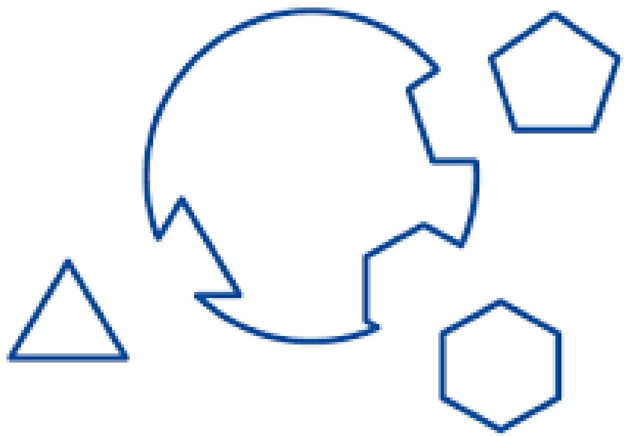
Did you know… carbonic anhydrase is one of the fastest enzymes known. Each enzyme molecule can hydrate 1,000,000 molecules of CO_2 per second.

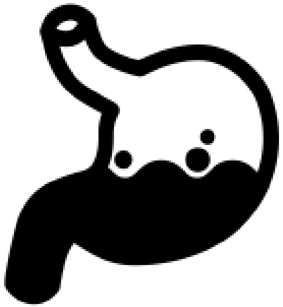
What are Enzymes?

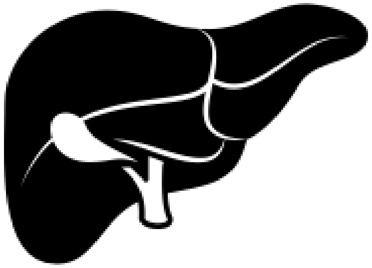
Enzymes are biological **catalysts** made from proteins.

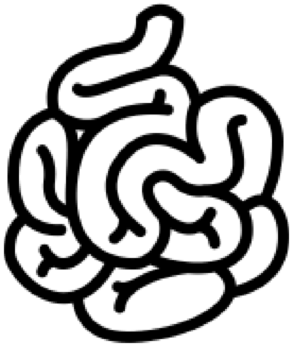
"A catalyst is something that speeds up a reaction"**.** Enzymes help reactions in our body happen faster.

Substrate: Molecules that enzymes act upon



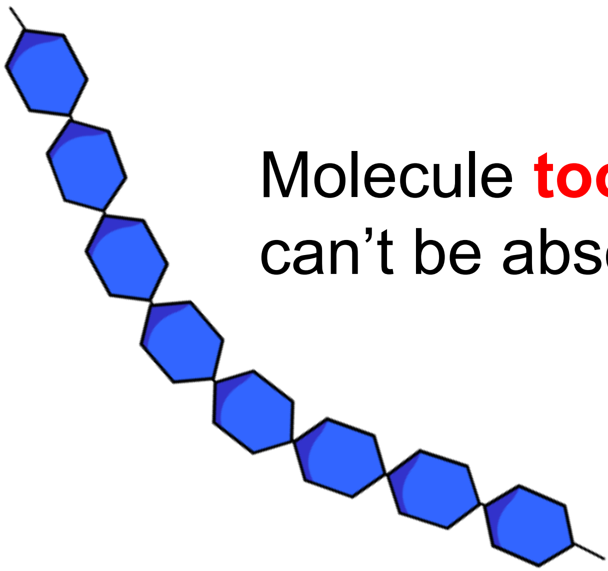






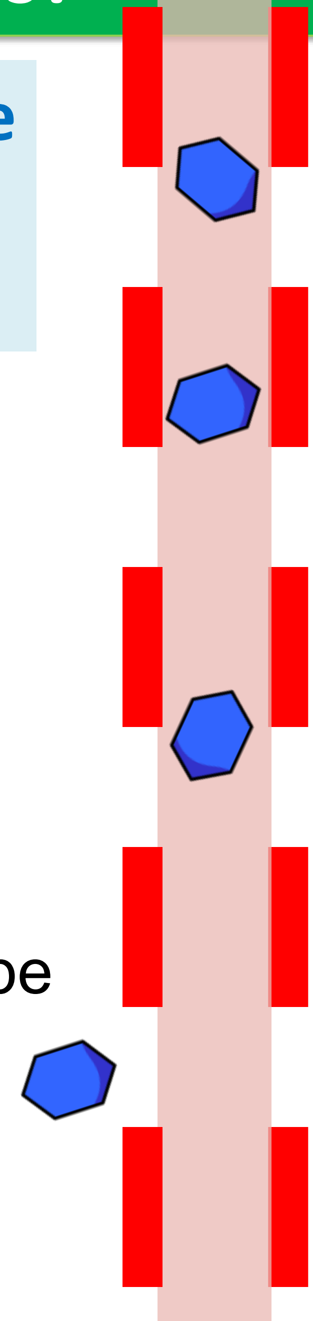
What is the role of digestive enzymes?

Digestive enzymes convert food into **small soluble molecules** that can be **absorbed** into the blood stream of the **small intestine.**



Molecule **too large…** can't be absorbed

Small enough to be absorbed.

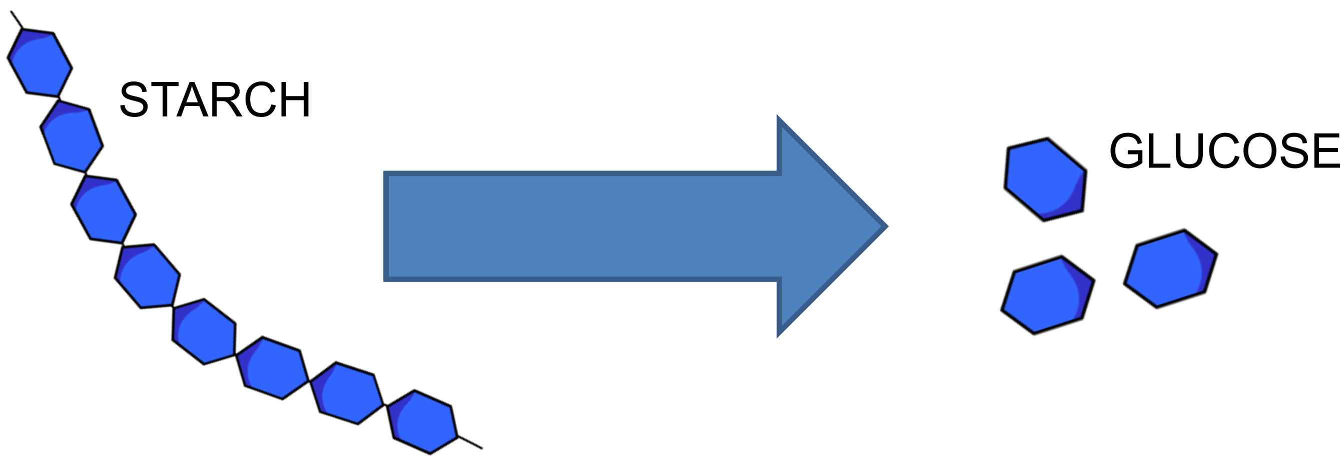


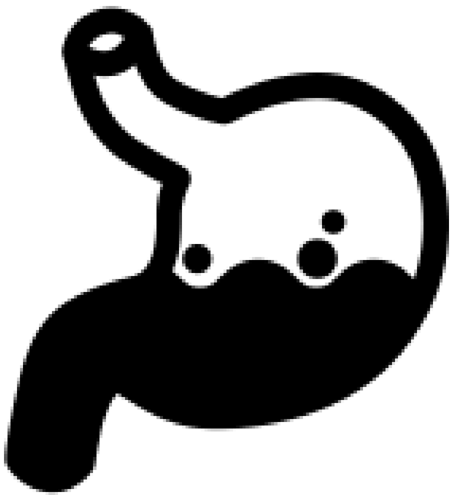
Organ → Mouth

Enzyme → Amylase

What does it break down → Carbonhydrate called starch into glucose

Substrate: Molecules that enzymes act upon



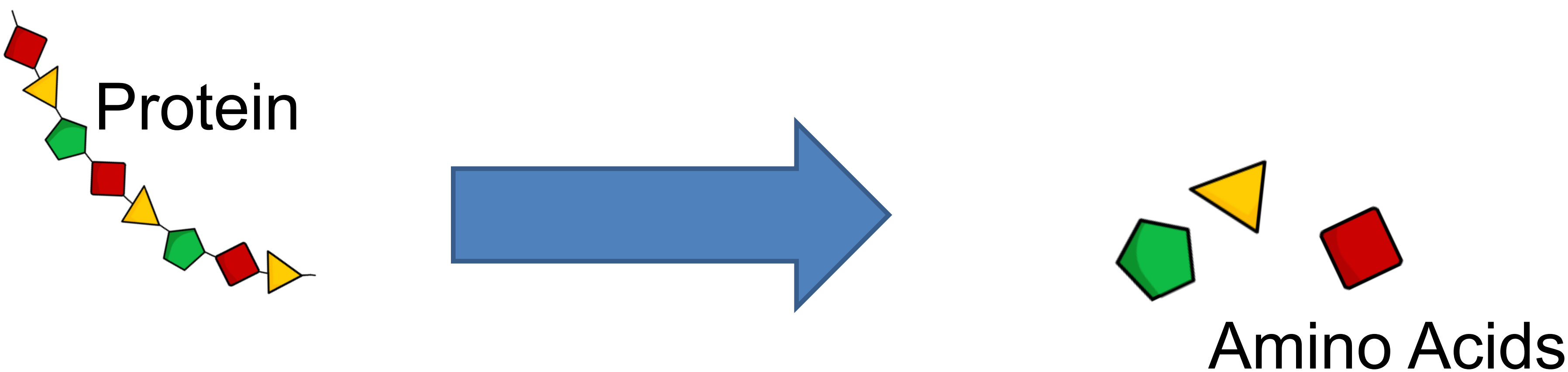


Organ → Stomach

Enzyme → Pepsin

What does it break down → Protein into Amino acids

What is the acid for → Help the enzymes Work better



Organ → Pancreas

Enzyme → Protease, lipase and carbohydrase

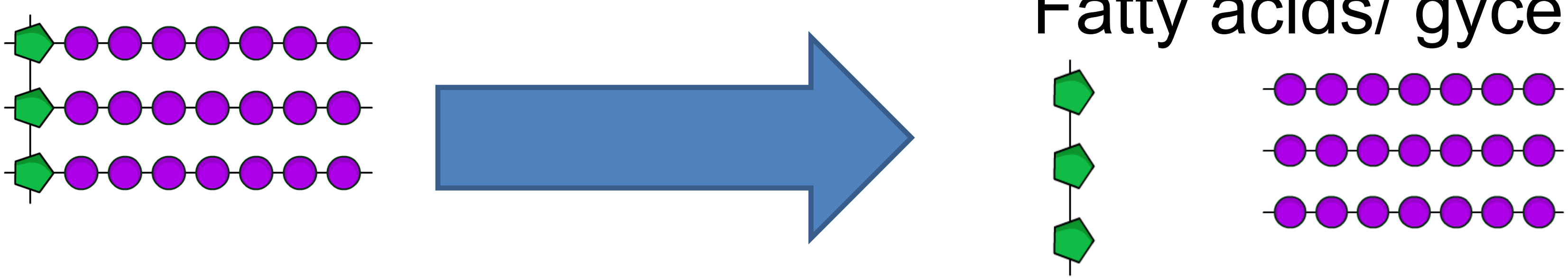
What does it break down → Protease breaks down protein

Lipase breaks down fat

Carbohydrase breaks Carbohydrate like starch

Fats

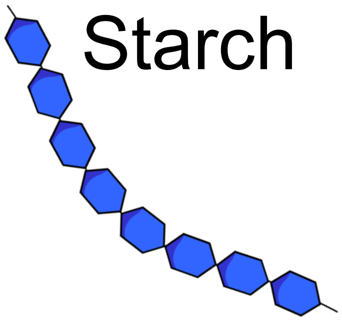
Fatty acids/ gycerol



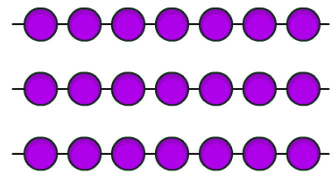
As we go through the following slides add into your sheet the missing information.

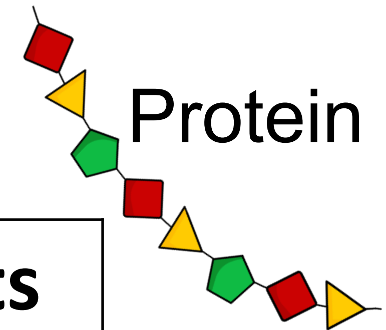
Substrate (draw a picture of it's structure)	Enzyme	Products

As we go through the following slides add into your sheet the missing information.



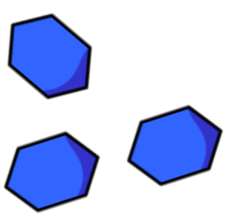
Starch



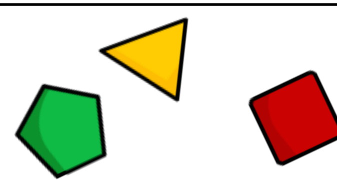


Protein

Substrate (draw a picture of it's structure)	**Enzyme**	**Products**
	_____ase	G______
P______		A____ A_



Glucose



Amino Acids

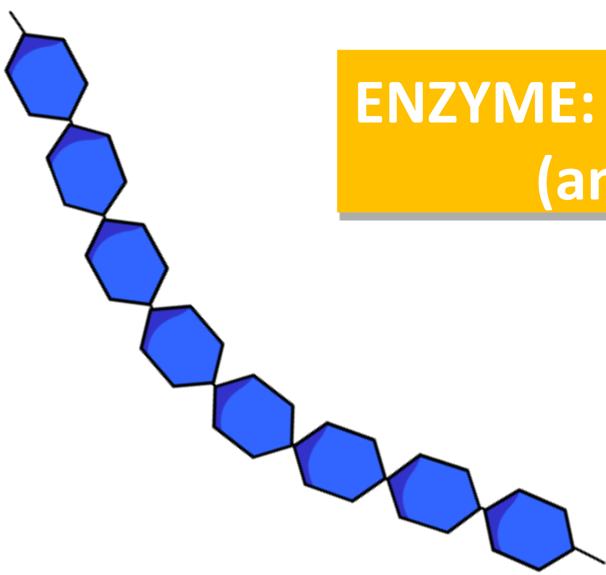
Carbohydrate digestion

Carbohydrates are chains of identical sugar molecules.

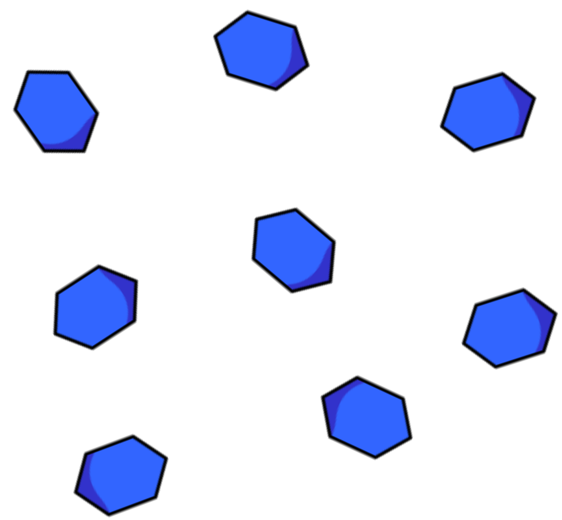
The digestive enzyme called **carbohydrase** breaks the chemical bonds between the individual sugar molecules in each carbohydrate chain. Digestion starts in the <u>mouth</u>.

SUBSTRATE
‘Starch’

PRODUCT
Simple sugars ‘Glucose’



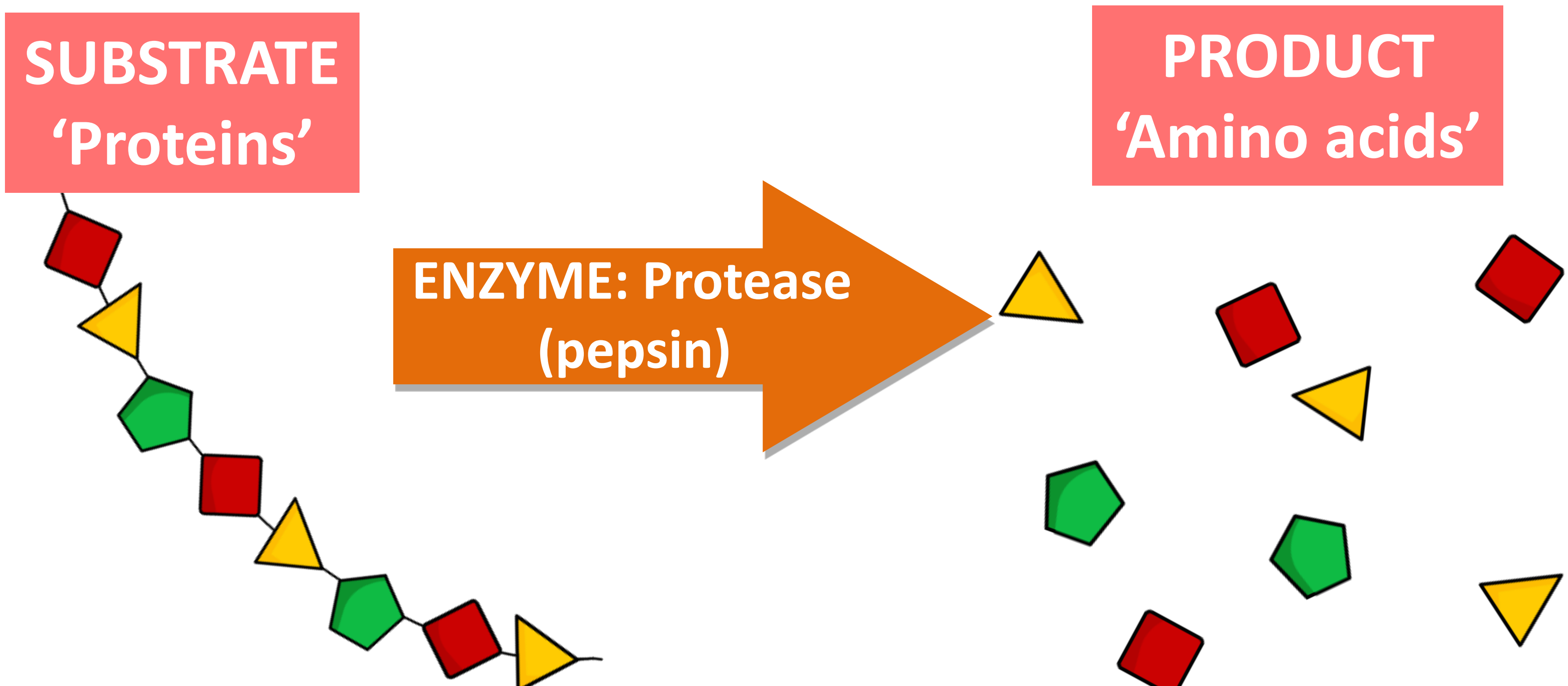
ENZYME: Carbohydrase (amylase)



Protein digestion

Proteins are made up of amino acids. There are over 20 different types of amino acids.

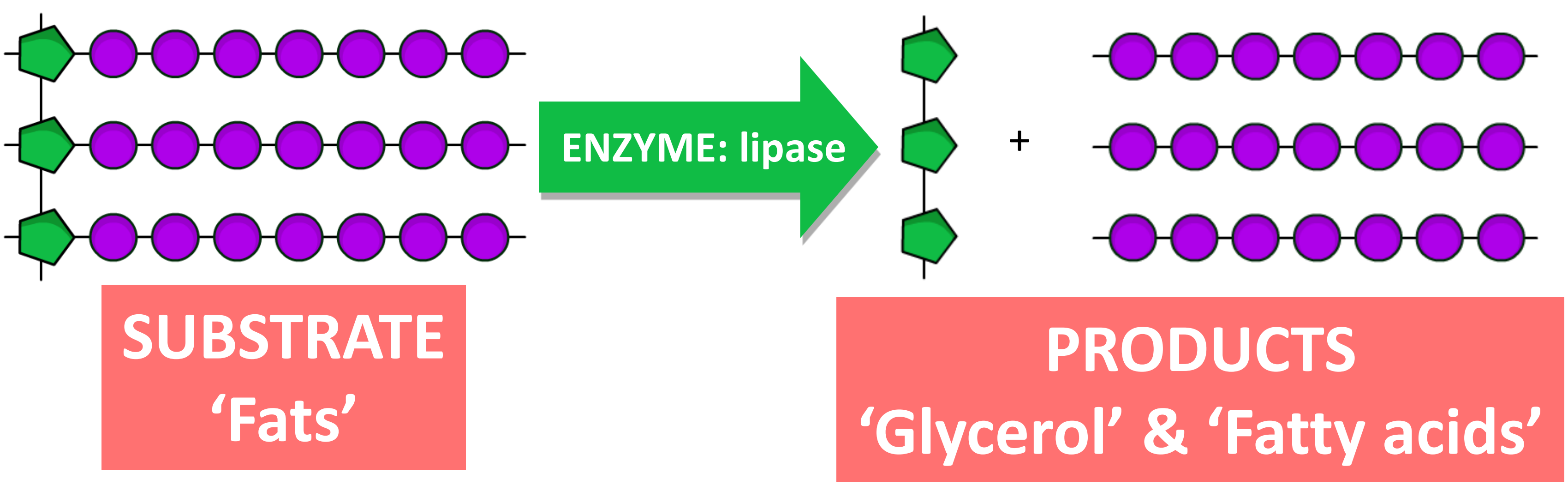
Proteins are digested by digestive enzymes called **proteases**. These enzymes work in the stomach and small intestine.



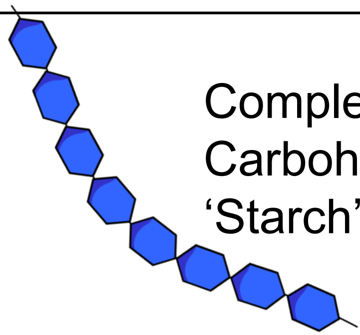
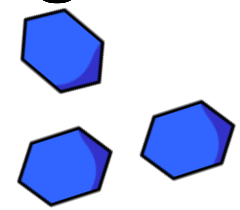
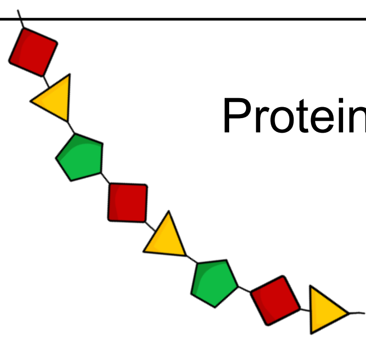
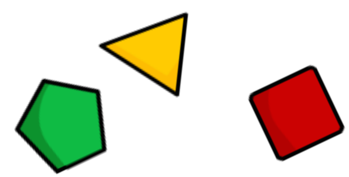
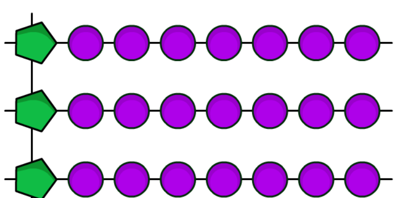
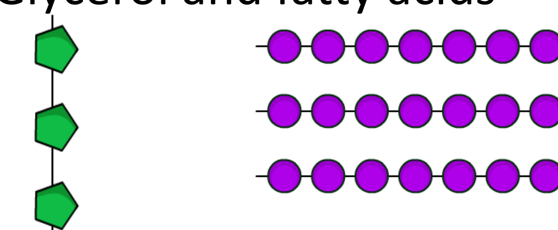
Two stage fat digestion in the small intestine

Firstly, **bile** (NOT an enzyme) allows the fat to 'mix' with water by breaking the fat into smaller droplets. This is called **emulsification**.

Secondly, the digestive enzyme **lipase** breaks each fat molecule into the smaller glycerol and fatty acid molecules.



As we go through the following slides add into your sheet the missing information

Substrate (draw a picture of it's structure)	**Enzyme**	**Products**
 Complex Carbohydrates 'Starch'	Carbohydrase 'Amylase'	Simple sugars Glucose 
 Proteins	Protease	Amino Acids 
Fats 	Lipase	Glycerol and fatty acids 

True

False

Bile is an enzyme?

Enzymes are a catalyst?

The acid in stomach breaks food down?

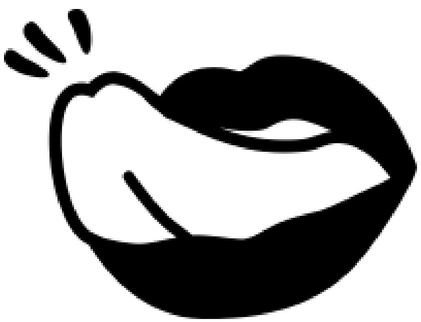
Enzymes are made from amino acids?

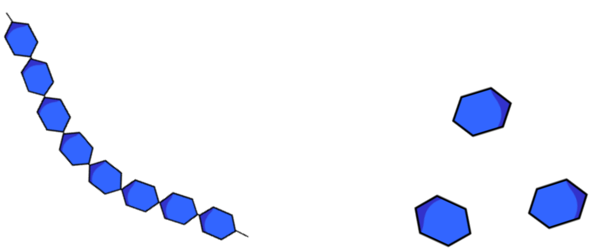
Catalyst decrease the speed of a reaction?

Enzymes are alive?

Digestive enzymes are found in these three organs

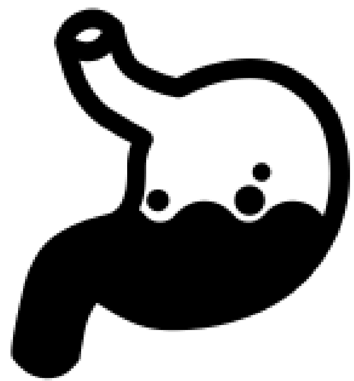
Amylase
carbohydrase
(Breaks down Carbohydrate)

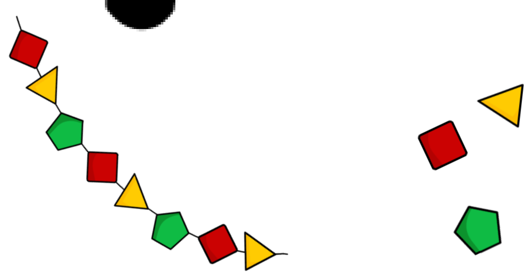




Starch → Glucose

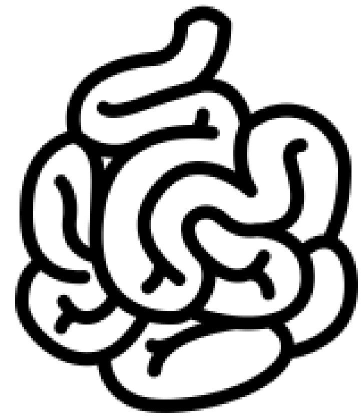
Pepsin
protease
(Breaks down Protein)

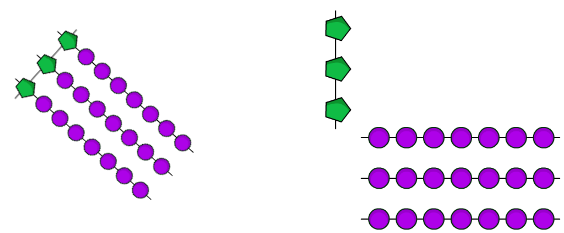




Protein → Amino acids

Lipase
(break down Lipids/fat)





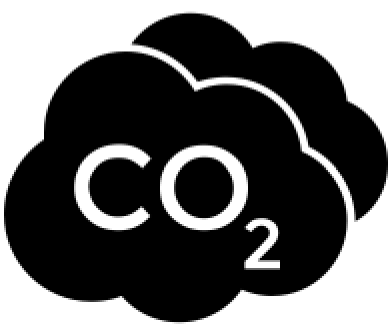
Fats → Fatty acids

LOb: Understand how the digestive system uses specific enzymes to absorb nutrients.

Keywords: Carbohydrase, amylase, protease, amino acid, lipase, glycerol, fatty acid, soluble, absorb

Learning Outcomes:

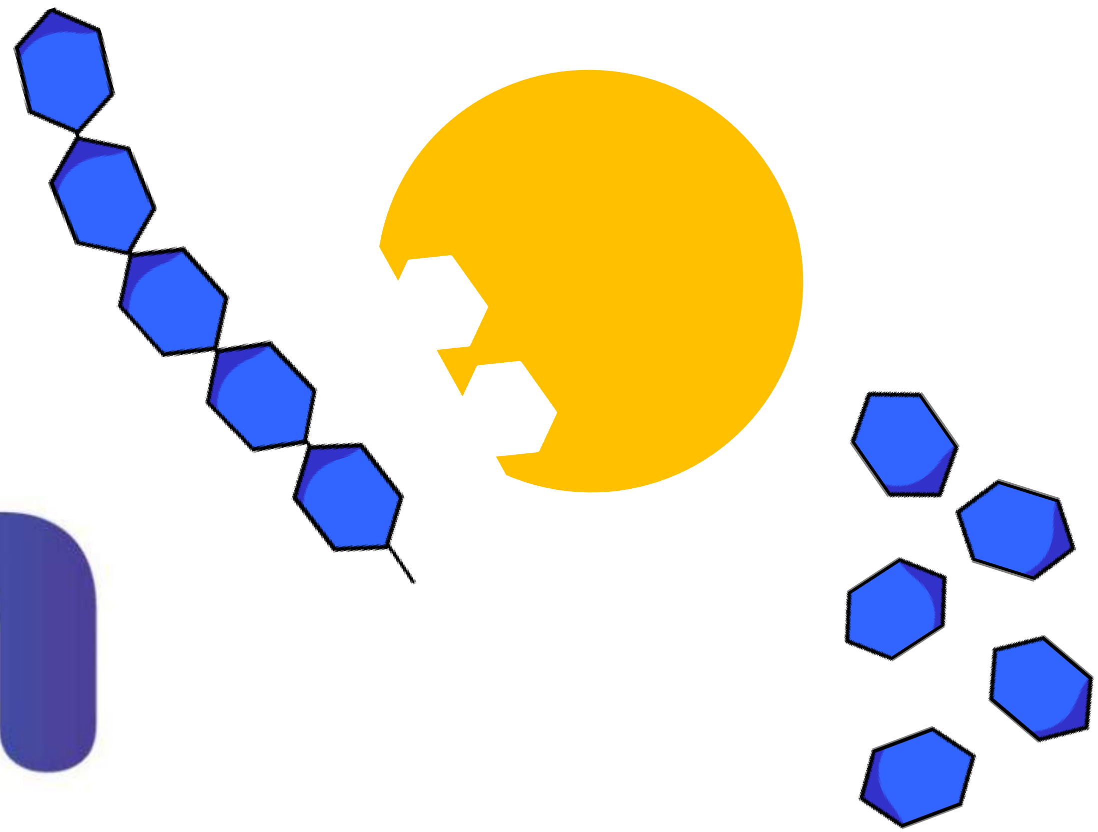
- ✓ Recall the digestive enzymes, where they are produced and what their substrate and product is
- ➢ Investigate the effect of pH on amylase enzyme



Did you know… carbonic anhydrase is one of the fastest enzymes known. Each enzyme molecule can hydrate 1,000,000 molecules of CO_2 per second.

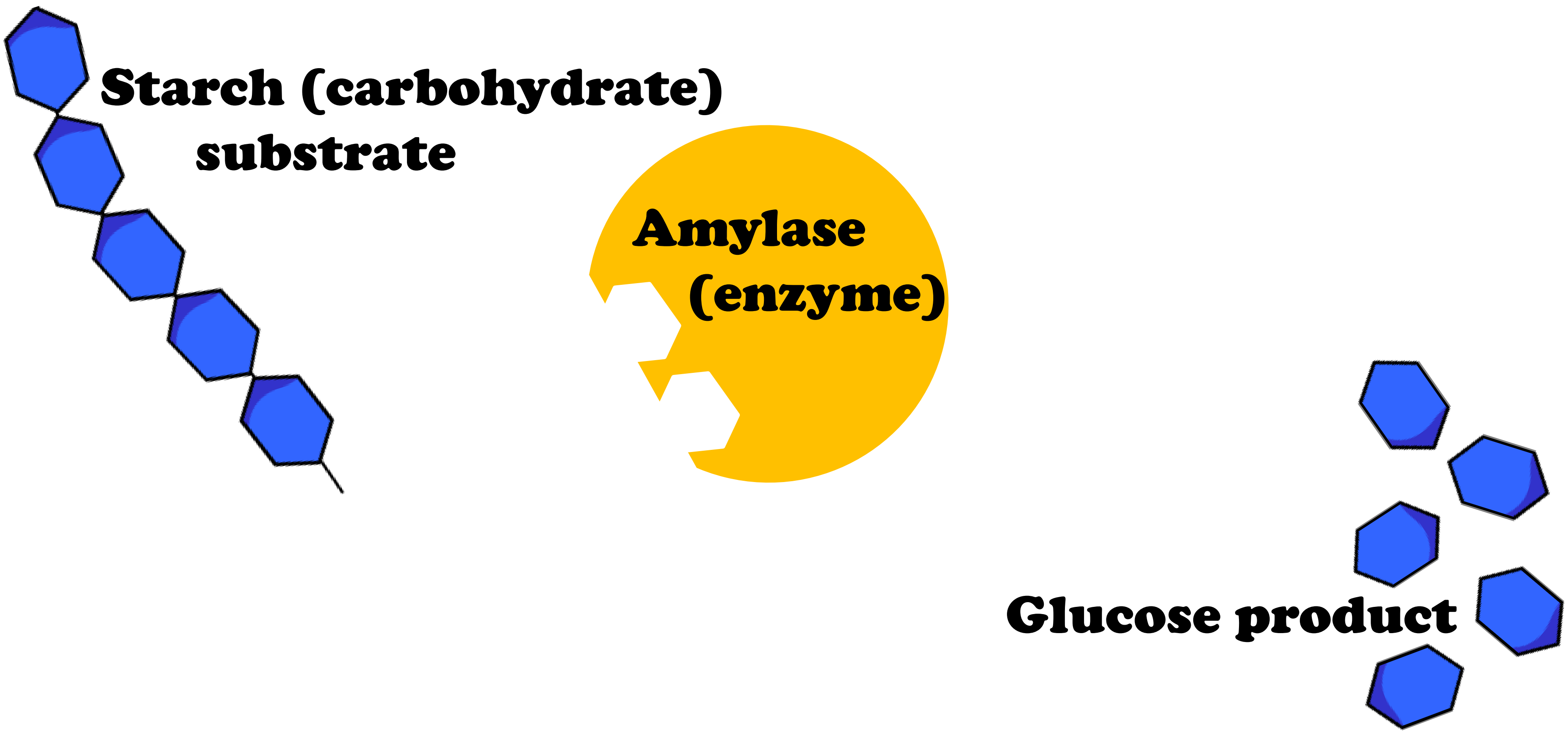
Experiment

"*Investigate the effect of* ***pH*** *on the* ***rate of reaction*** *of* ***amylase enzyme.***"



Background

We want to see whether different pH solution increase or decrease the ability of amylase to turn starch into glucose. To do this **we need something that tells us when starch is present** and when it is no longer there.

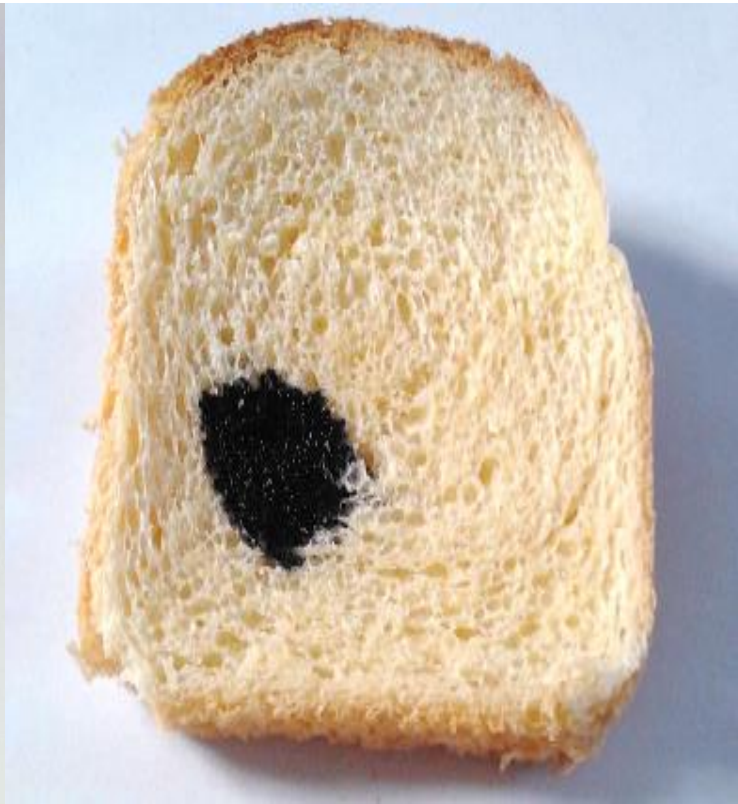


Test for starch – Iodine indicator solution

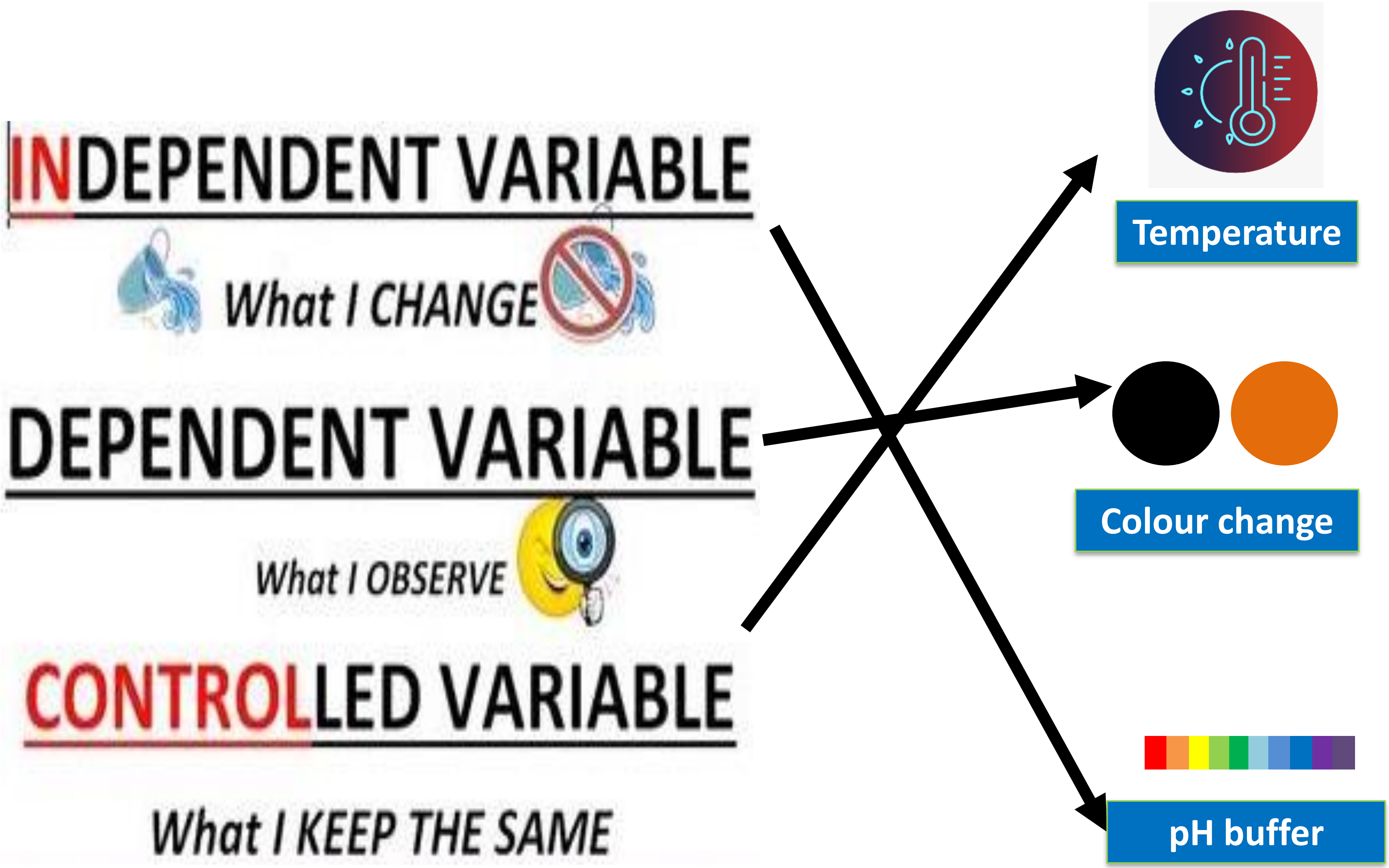
In the presence of starch iodine turns black.
When starch is not present iodine is orange.

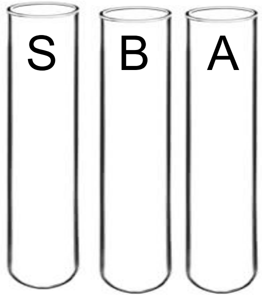
Starch

No Starch



Can we match the variable to the pictures?





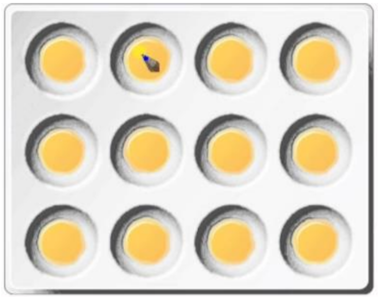
Label three test tubes.
One with the letter S
One with the letter B
One with the letter A
Place into a water bath
Keep as close to 35°

2

Using a three separate pipettes
Place 2 cm^3 of starch solution
Into tube S
Add 2 cm^3 of buffer solution **pH 5**
into tube B
Add 1cm^3 of amylase solution
Into tube A

3

When the solutions
are at 36° you are
ready to start



4

Add a few drops of
iodine solution into each
spot of the spotting tile.
Then transfer the
contents of test tube S
and B into tube A

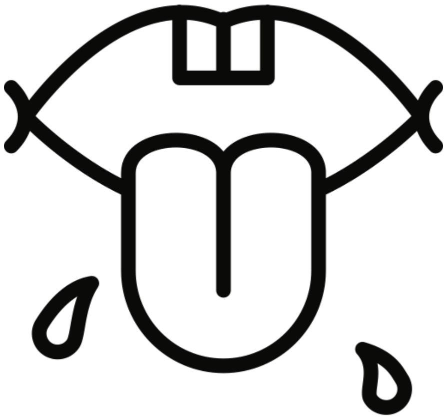
Quickly take a pipette
and add a drop
of the mixture in tube A
into the first dimple
starting the timer. Repeat
every 10s

5

Stop once the colour is orange
and doesn't change. Record the
time by counting the number
of spots till orange.
Repeat with pH 6, 6.4, 7 and 8

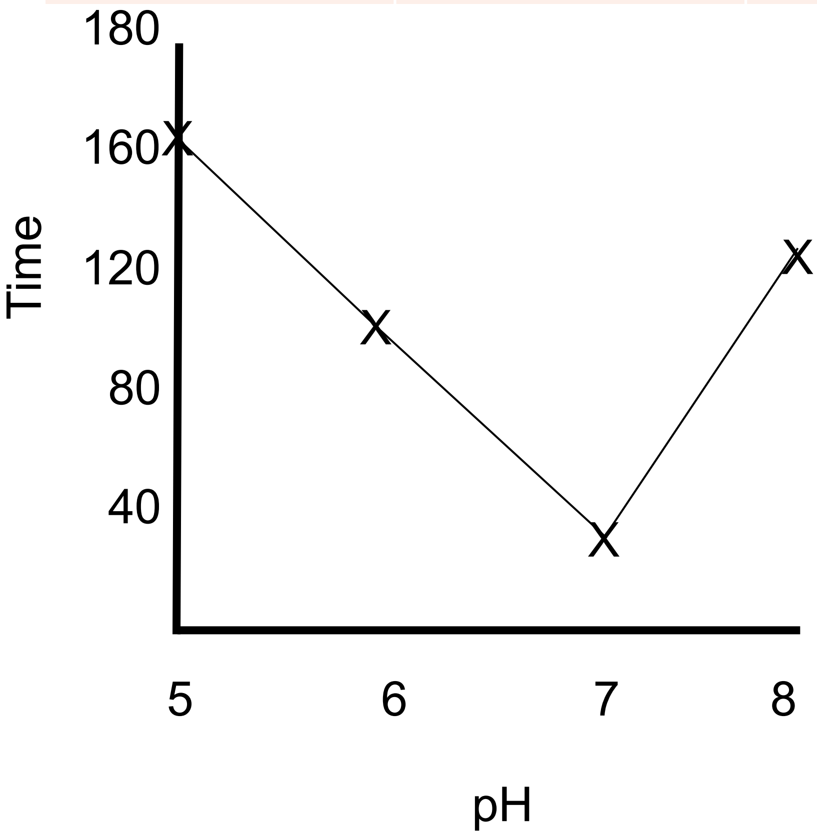
Saliva is between pH6-7. Copy table and complete the means results.

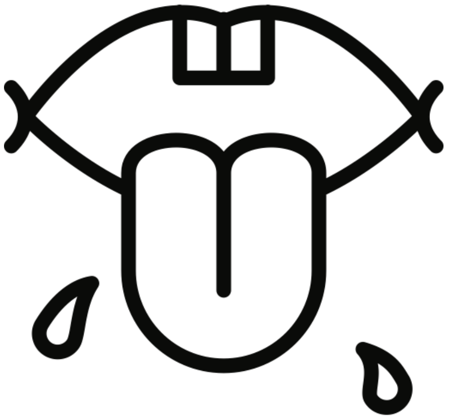
pH	Test 1 (sec)	Test 2 (sec)	Test 3 (sec)	Mean
5	170	150	170	
6	110	100	100	
7	30	40	30	
8	120	130	180	



Review

pH	Test 1 (sec)	Test 2 (sec)	Test 3 (sec)	Mean
5	170	150	170	163
6	110	100	100	103
7	30	40	30	33
8	120	130	180	125





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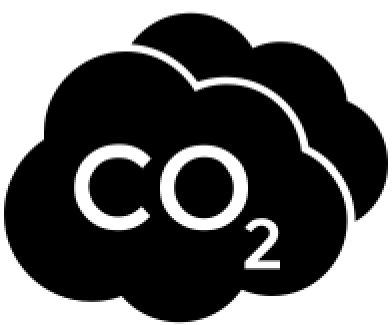

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keyfacts

- **Factors** that effect enzymes: **pH, temperature, substrate and enzyme concentration, pressure, surface area**

- Each enzyme has an **optimal condition.**

- **Temperature** effects enzyme activity. **Cold temp = low kinetic energy reduced frequency of collisions.** .

- **Denaturing** at **high temperature** or a **pH range** means that the **structure of proteins** has been **irreparably damaged** which **stops the enzyme** from being able to **catalyse** a reaction.

- **H^+ ions exert forces** on enzyme protein structures which are either too strong or to weak to what they are used to so they denature.

Plenary: Chose a box

7. If I had to set a test based on today's lesson, I would ask the question:

4. With the information that I learnt today, I will now be able to ...